Model Training and Hyperparameters Report

Robust Z-scores (Alisa's Progress)

For the baseline, I used robust z-scores calculated within peer groups. I compared minimum group sizes of 30, 50, and 100 with the 3-level fallback. The overall flag rate barely changed (1.21% to 1.24%), and 88.9% of flags overlapped between 30 and 100. However, rows in broader groups were flagged more often: around 1.1% at Level 1, 3% at Level 2, and 13% at Level 3. At size 100, about 271K rows moved from Level 1 to broader groups, and their flag rate increased from 2.01% to 3.22%.

Table 1: Provider types affected by moving to broader groups.

Provider Type	Moved Rows	Flagged Only at 100
Ambulatory Surgical Center	5,016	36.3%
Portable X-Ray Supplier	1,593	11.5%
Geriatric Medicine	2,326	7.0%
Diagnostic Radiology	9,356	5.5%

Table 2: Comparison of flagging rules.

Rule	Flag Rate
Any 1 feature > 3.5	5.91%
2+ features > 3.5	0.16%
Our rule	1.24%

The biggest difference was for Ambulatory Surgical Centers. At Level 2 we group only by HCPCS and POS, so different provider types can end up in the same group. Surgical centers have facility fees, so their payments can be higher than for physicians even if nothing unusual is happening. If we use 100 as the minimum group size, we need to mark these flags in the results, since they might be caused by comparing different types of providers.

For flagging, I used 1 feature > 10 OR 2+ features > 3.5. I chose 3.5 since it is a commonly used threshold for modified z-scores. However, using it for any single feature would flag 5.91% of rows (around 580K), which is too many to review. Requiring 2+ features > 3.5 would flag only 0.16%, since our features have low correlations and are not often high at the same time.

I also tested different thresholds for a single feature. With 8, the flag rate was 1.60%, and with 12 it was 1.03%. The rows we would lose at 12 had around 1.38 times more services per patient than their peers, which is similar to the rows we keep. At 8, we would add more rows with relatively small differences, like payments only 4% above peers. So I decided to keep 10, which flags 1.24% of rows.

Isolation Forest (Diana's Progress)

Feature experiments

A total of 11 columns were considered during experiments: total beneficiaries, total services, total beneficiary-day services, services per beneficiary, services per beneficiary-day, average submitted charge, average Medicare allowed amount, average Medicare payment amount, average Medicare standardized amount, charge-to-allowed ratio, and payment-to-allowed ratio.

Feature selection was tested using several methods. First, within-peer variation was examined to determine whether each feature changed enough to help Isolation Forest distinguish unusual records. `Services_per_bene_day` showed very little variation, with a median IQR near 0 and constant values in about 22% of tested peers. Spearman correlation was then used to identify redundant features. Three raw volume measures were extremely correlated, with correlations between about 0.985 and 1.00, while submitted charge and charge to allowed had a correlation of about 0.95. SHAP was used to measure which features influenced the Isolation Forest anomaly scores, while leaving one feature out ablation tested how much rankings changed when each feature was removed.

Based on these experiments, the final feature set proposed was reduced to 5 variables: services per beneficiary, charge / allowed, total beneficiaries, payment / allowed, and average Medicare standardized amount. After this assignment's submission, post-training feature importance will be tested.

Hyperparameter experiments

Tests showed that increasing the number of trees consistently improves Isolation Forest stability. Although 400 trees produced the highest stability (Spearman = 0.990 and Top-5% Jaccard = 0.886), its average fit time increased to 0.385 compared to 0.190 seconds for 200 trees, while the additional stability improvement was relatively small. Changing `max_samples` produced only minor differences, so `auto` was retained, while using all four features (`max_features` = 1.0) gave the strongest overall ranking stability. **Based on the balance between stability and computational cost, 200 trees,** `max_samples` **= auto, and** `max_features` **= 1.0 were selected as the final setting.**

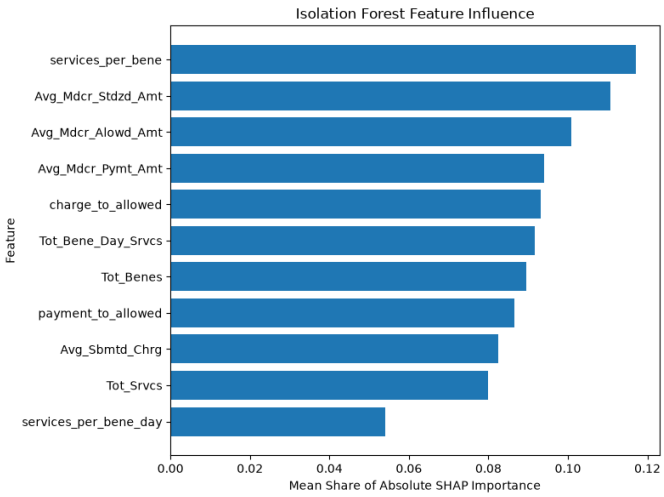


Figure 1: Isolation Forest feature influence (mean share of absolute SHAP importance).

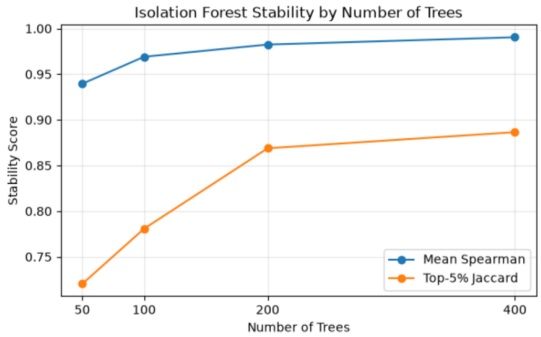


Figure 2: Isolation Forest stability by number of trees (mean Spearman and Top-5% Jaccard).

DBSCAN (Tyson's Progress)

Feature experiments

Since DBSCAN requires groups of categorical columns as input, a total of 5 features were tested: HCPCS Code (1), Place of Service (2), Provider Type (3), Provider Entity Code (4), and State Abbreviation (5). I tested 6 combinations of these features: 12, 123, 1234, 1235, 1245, and 12345. The results of this testing provided insights into which features should be kept or discarded, and which groups were best for our particular use case. **Group 123 is the best for accurate general analysis among the data, while 1235 is best for including geolocation within the groups. Feature 4 skews outliers, needlessly splits groups, and actually removes useful information rather than gaining information. Feature group 12 has too big of groups to be granular enough to make good accurate predictions of outliers and anomalies.**

Hyperparameter experiments

Looking into the two groups (123 and 1235), I tested out the `eps` hyperparameter. This hyperparameter changes the minimum distance required to be included into the group. Within each graph of the peer groups there is an elbow where the outliers start to be excluded; having the red `eps` line right in the middle of that elbow is the desired outcome. I tested values from 0.4 (very restrictive) to 1.1 (very inclusive), going up by 0.1 each test for each group. **For group 123, the results showed that even though there are outliers like `peergroup_005` where the best `eps` is 0.5, the best overall `eps` value is 0.9. This seems to hit the elbow right where we need most of the time. For group 1235 the overall best `eps` is 1.0. Since the groups are similar to 123, just with added geolocation, it makes sense that we need to be slightly more inclusive compared to 123 since the data has gotten smaller and each record has fewer records to compare to.**

Feature Experiments and Conclusion (Alisa's Progress)

I compared 3 feature sets: all 7 raw features, 4 selected raw features, and our 4 engineered features. The problem with raw features is that some of them carry the same information, so the model can flag the same reason multiple times.

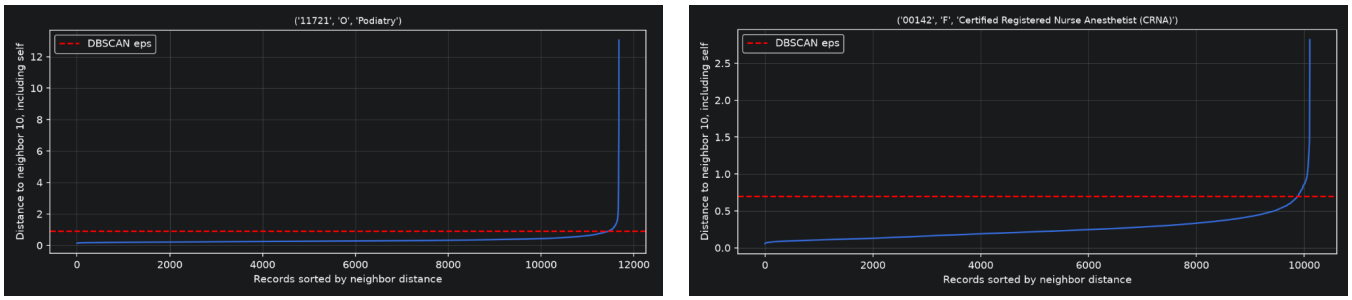


Figure 3: Sorted 10th-neighbor distance with the DBSCAN `eps` value (red dashed line) for two peer groups: `00142`, F, Certified Registered Nurse Anesthetist (left) and `11721`, O, Podiatry (right).

Table 3: Feature set comparison.

Feature Set	Flag Rate	Double-counted Flags
A: 7 raw	0.99%	70.75%
B: 4 raw	0.60%	53.89%
C: 2 raw + 2 engineered	1.24%	0.00%

For sets A and B, 71% and 54% of flags were double-counted. Set C had 0% under our definition, since each feature was treated as a separate category. The overlap between the top 1% anomalies from C and the raw sets was only around 28%, so most of the rows identified by our features were different.

I also tested adding `payment_to_allowed` as a fifth feature. It added 51,306 new flags (29.7% of all flags). In 86% of these cases, `payment_to_allowed` alone had a robust z-score > 10, while the median difference from the peer median was only 0.8 percentage points. Medicare usually pays around 80% of the allowed amount, or 100% for some services like vaccines. Within peer groups this value barely changes, so even small differences can give very high z-scores. Most of these differences are related to Medicare payment rules rather than unusual provider behavior.

Conclusion

The results were similar to those from the Isolation Forest experiments, where almost the same features were selected, except for `payment_to_allowed`, which was excluded after further analysis. Therefore, we decided to keep: standardized payment, beneficiaries, services per beneficiary, and markup. They have low correlations with each other, and all models gave similar results.